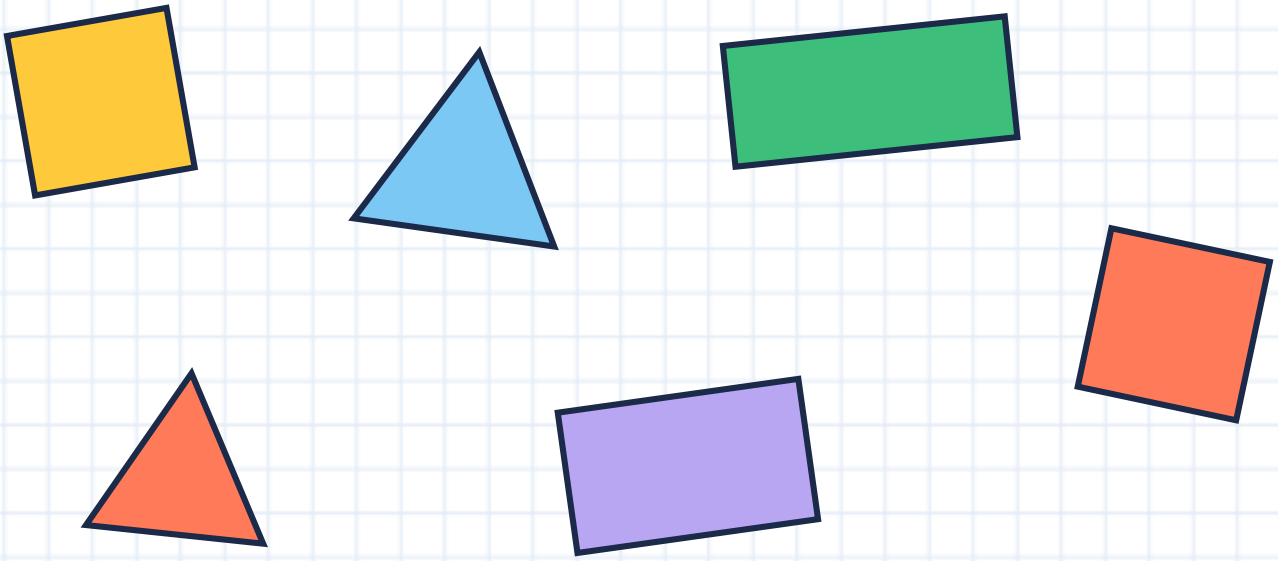


LEVEL 2 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Digit Detectives

Lesson 2
Missing Digit Mysteries

Lesson 3
Legs, Wheels and Arrays

Lesson 4
Pattern Machines

Lesson 5
Mystery Number Machines

Lesson 6
Market Day

Lesson 7
Fair Shares

Lesson 8
Shape Builders

Lesson 9
Measure Masters

Lesson 10
Challenge Day

Tutor: _____ Program dates: _____

START HERE. After question 3 you may choose: carry on, or try Think harder.

DIGIT DETECTIVES

Name _____ Group _____ Date _____

A 3-digit number has hundreds, tens and ones. Where a digit sits tells you what it is worth.

No.1

Place value. Fill in the blanks.

a 4 hundreds, 0 tens and 7 ones = _____

b 6 hundreds, 15 tens and 2 ones = _____

c 830 = 8 hundreds and _____ tens 830 = 7 hundreds and _____ tens

d In 572, the digit 7 is worth _____

No.2

Use the digit cards 2, 7 and 4. Use each card once to make a 3-digit number.



a Make a list in order. Write every number you can make.

Start with 2	Start with 4	Start with 7
_____ _____	_____ _____	_____ _____

b Biggest number: _____ Smallest number: _____

c Which of your numbers is closest to 500? _____

d How many of your numbers are odd? _____

No.3

Now use the digit cards 0, 5 and 9.



a List every 3-digit number you can make, in order.

Numbers: _____

b The cards 2, 7, 4 made 6 numbers. Why do 0, 5, 9 make fewer?

.....

.....

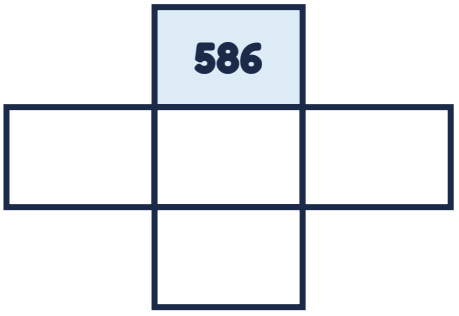
No.4

These are pieces cut from a 1000 chart. Each row has 10 numbers. Fill in the empty squares.

Piece A

347		
	358	
		369

Piece B



a On the chart, one square DOWN is _____ more. One square RIGHT is _____ more.

b Which number is 2 squares below 385 on the chart? _____

No.5

Look for the pattern. Fill in the next numbers.

a 47, 147, 247, _____, _____

b 532, 522, 512, _____, _____

c 980, 985, 990, 995, _____

d 703, 702, 701, _____, _____

No.6

Digit riddles. Who am I?

a I am a 3-digit number. My hundreds digit is 4. My ones digit is 2 more than my tens digit. My digits add up to 12.

I am: _____

b I am a 3-digit number. My digits add up to 3. Find ALL the numbers I could be. Tip: use 3 counters. Put 1 counter in the hundreds box, then share the other 2 between the tens and the ones. Then try 2 counters in the hundreds box.

How many numbers? _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Mirror numbers. A mirror number reads the same forwards and backwards, like 252 or 909. (a) Write 5 mirror numbers of your own. (b) Find ALL the mirror numbers from 200 to 299. How do you know you have found them all? (c) Ali says every mirror number is odd. Is he right?

Easy first step: write 252, 909 and 131, then write three more mirror numbers.

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ made a list in order ☐ used digit cards or counters ☐ I asked for help ☐ I am still stuck

(my tutor will read this at the next lesson)

Story corner: Aryabhata India, 476 - about 550 (sources differ)

Aryabhata lived in India about 1,500 years ago and wrote a short maths book in verses. One rule in his book says each place in a number is worth ten times the place before it. That is the idea behind hundreds, tens and ones.

Try it (if you like): Aryabhata's rule: each place is ten times the place before it. In the number 538, which digit is in the tens place, and what is it worth? What is ten times 50?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.

No.2

Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

No.3

What number is 3 hundreds, 14 tens and 6 ones?

No.4

Mei says: "If I swap the tens digit and the ones digit of a 3-digit number, the new number is always smaller." Is she right? Show what happens with some examples. When does the number stay the same?

Go further

No.5

Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?

No.6

Ali counts in 5s, starting at 3: 3, 8, 13, 18 ... Will he say 500? Will he say 998? Explain.

No.7

Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?

START HERE. After question 3 you may choose: carry on, or try Think harder.

MISSING DIGIT MYSTERIES

Name _____ Group _____ Date _____

Each yellow box hides one digit (on the slides a hidden digit is a letter such as A or B). Start at the ones column. Ask: was there a carry?

No.1

Find the missing digits. Write them in the boxes.

(a)

$$\begin{array}{r} 2\square 4 \\ +\ 35\square \\ \hline 579 \end{array}$$

(b)

$$\begin{array}{r} 46\square \\ +\ \square 75 \\ \hline 837 \end{array}$$

(c)

$$\begin{array}{r} \square 86 \\ +\ 2\square 9 \\ \hline 645 \end{array}$$

Check one of your answers by adding the full numbers: _____ + _____ = _____

No.2

Find the missing digits. Try a digit and check it. Check: answer + the number you took away = the number you started with. Tip for (b): 3 – 7 cannot be done, so think 13 – 7.

(a)

$$\begin{array}{r} 7\square 8 \\ -\ 32\square \\ \hline 453 \end{array}$$

(b)

$$\begin{array}{r} 5\square 3 \\ -\ \square 67 \\ \hline 276 \end{array}$$

No.3

The = sign means "is the same as". Fill in the boxes.

a 350 + 140 = 300 + _____

b 625 – 200 = _____ – 100

c 480 + _____ = 500 + 260

d Ravi says 398 + 205 = 400 + 203 is true. Is he right? Explain how you know without adding.

No.4

Magic square. Every row, column and diagonal adds up to the same total.

80		60
	50	
40		20

a What is the magic total? (Find a line with all three numbers.) _____

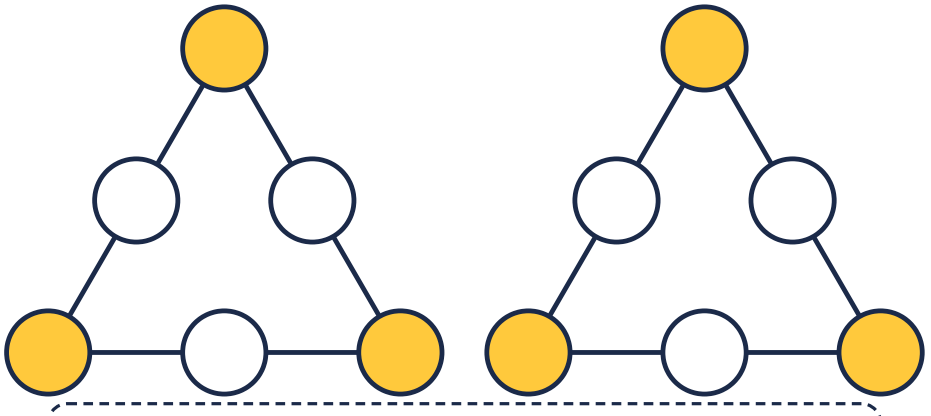
b Fill in the empty squares.

c Add 100 to every number. Is it still a magic square? What is the new total? Explain.

No.5

Magic triangle. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so every side has the same total.

(a) Each side = 390 **(b) Each side = 420**



Number tiles: use all six in EACH triangle

110	120	130	140	150	160

a Each side adds up to 390. Write the numbers in the circles. Which numbers are in the corners? _____ (Start: try 110 and 120 as two corners. What goes between them?)

No.6

Mei thinks of two numbers. They add up to 700. One number is 200 more than the other. What are the two numbers? Draw a model: two bars, and the long bar is 200 longer.

Answer: _____ and _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Odd or even?. Ali says: "When I add two odd numbers, I always get an even number." (a) Test Ali's idea with 5 pairs of odd 3-digit numbers (for example 345 + 271). (b) Now test odd + even, and even + even. What do you notice? (c) Explain WHY, using pairs of counters or a drawing.

Easy first step: use counters to try 3 + 5, 7 + 9 and 11 + 13. Are the answers odd or even?

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ tried a digit and checked it ☐ drew it ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Carl Friedrich Gauss Germany, 1777–1855

Gauss was a German mathematician. A story often told says that, as a boy, he added 1 to 100 fast by pairing 1 with 100, 2 with 99 and so on (historians are not sure it happened exactly like this). Pairing numbers to add faster is real, clever maths.

Try it (if you like): Use Gauss's pairing trick to add 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10. Which pairs do you make?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

A magic triangle has 3 corner circles and 3 circles in the middle of the sides. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so that every side adds up to 420. Draw the triangle. Which numbers are in the corners?

No.2

Find the missing digits: 5A7 + B48 = 905

No.3

Find the missing digits: 7A1 – B58 = 263

No.4

What number makes this true? 538 + 199 = 540 + A

Go further

No.5

Find ALL the ways to fill in the missing digits: AB7 + 4CD = 1000

No.6

Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?

No.7

Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.

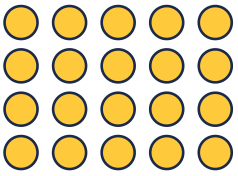
START HERE. After question 3 you may choose: carry on, or try Think harder.

LEGS, WHEELS AND ARRAYS

Name _____ Group _____ Date _____

No.1

Mei puts counters in an array.

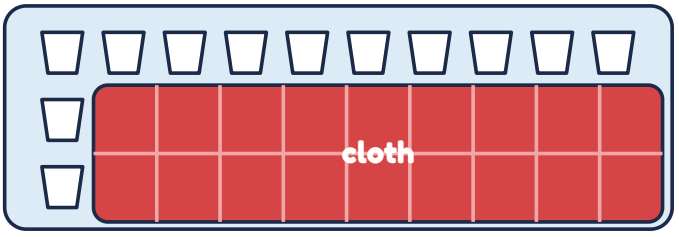


a Two multiplication sentences: _____ × _____ = _____ _____ × _____ = _____

b Two division sentences: _____ ÷ _____ = _____ _____ ÷ _____ = _____

No.2

A tray has the same number of cups in every row. A cloth covers some of the cups.



a How many cups altogether? _____ (b) How many under the cloth? _____

c How did you work it out without lifting the cloth?

No.3

A pen has hens and goats. A hen has 2 legs and a goat has 4 legs. There are 7 animals and 20 legs altogether. Start with all hens. Then swap one hen for a goat each time.

Hens	7				
Goats	0				
Legs	14				

a How many hens and how many goats? _____ hens _____ goats

No.4

Ravi has 23 stickers. He shares them equally among 4 friends, giving each friend as many as he can. Tip: draw 4 friends and deal out the stickers one at a time.

a How many does each friend get? _____ (b) How many are left over? _____

c What is the smallest number of extra stickers Ravi needs so that none are left over? _____

No.5

28 children go on a trip. Each van holds 5 children.

a How many vans are needed? _____ (b) How many empty seats? _____

No.6

Mr Tan puts 20 chairs in equal rows. There must be at least 2 rows and at least 2 chairs in each row. Find ALL the ways. Try 2 rows, 3 rows, 4 rows and so on. Stop at 10 rows.

Rows	2	3	4	5	6	7	8	9	10
Chairs in each row									
Does it work?									

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Where tables meet. (a) List the numbers up to 50 that are in BOTH the 2 times table and the 5 times table. What do you notice about them? (b) Now find the numbers up to 50 that are in both the 3 and 5 tables. (c) Is every number in both the 2 and 5 tables also in the 10 table? Explain.

Easy first step: write the 5 times table up to 50, then circle the even numbers.

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Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ made a table ☐ drew arrays or used counters ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Shakuntala Devi India, 1929–2013

Shakuntala Devi was born in Bangalore, India, in 1929. She was so quick at sums that in 1980 she multiplied two 13-digit numbers in her head in 28 seconds, which went into the Guinness Book of World Records. People called her the human computer.

Try it (if you like): Do these in your head like Shakuntala: 7 × 5, 9 × 4 and 6 × 10. Which answer is the biggest, and which is the smallest?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Find ALL the ways. Start with 1 tricycle, then 2 tricycles, and so on. Make a table or use counters. How do you know you have found them all?



bicycle: 2 wheels



tricycle: 3 wheels

No.2

There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?

No.3

Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

No.4

An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?

Go further

No.5

A toy shop has tricycles (3 wheels) and toy cars (4 wheels), at least one of each. There are 30 wheels altogether. Find all the ways.

No.6

Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?

No.7

A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?

START HERE. After question 3 you may choose: carry on, or try Think harder.

PATTERN MACHINES

Name _____ Group _____ Date _____

No.1

This pattern keeps going in the same way. Here are places 1 to 10. Tip: draw a line after the LAST shape of each group.



1 2 3 4 5 6 7 8 9 10

place

a What shape is in place 20? _____ (b) In place 23? _____

c What shape is in place 41? _____ How do you know?

No.2

Ali writes the letters M, A, T, H again and again.



1 2 3 4 5 6 7 8

a What is the 18th letter? _____ (b) Where is the 10th letter H? place _____

c Ali writes 30 letters. How many Ts does he write? _____ Show how you know.

No.3

Siti's bead string goes 2 red, 3 yellow, 2 red, 3 yellow ... It has 12 red beads and ends with yellow beads. Tip: draw one group, then count the groups.



a How many yellow beads? _____ (b) How many beads altogether? _____

No.4

Each flower has 5 petals.



a Fill in the table.

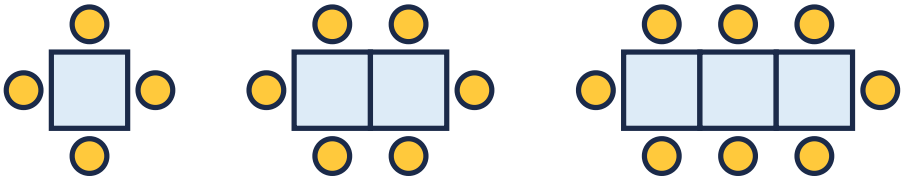
Flowers	1	2	3	4	5	6
Petals	5					

b Petals on 10 flowers? _____ (c) 35 petals make how many flowers? _____

d Mei has 23 petals. How many flowers? _____ Petals left over? _____

No.5

Square tables are pushed together in a row. One child sits at each open side.



1 table **2 tables** **3 tables**

a Fill in the table.

Tables	1	2	3	4	5	6
Children	4	6				

b Children at 10 tables? _____ (c) Tables for 20 children? _____

d Why does the number of children go up by 2 each time?

No.6

Ali counts in 3s: 3, 6, 9 ... Mei counts in 4s: 4, 8, 12 ... Which numbers from 1 to 36 do they BOTH say? Make two lists.

a Ali: 3, 6, 9, _____

b Mei: 4, 8, 12, _____

c Both say: _____ What do you notice? _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Three counting patterns. Pattern A: 1, 4, 7, 10 … (add 3 each time). Pattern B: 2, 5, 8, 11 … Pattern C: 3, 6, 9, 12 … (a) Write each pattern up to 30. (b) Pick any number from 1 to 30. Which pattern is it in? Can a number be in two patterns? Can a number be in none? (c) Why does this happen?

Easy first step: write pattern A, 1, 4, 7, 10 …, up to 30.

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was … .....

What helped me: ☐ counted in groups ☐ wrote the pattern out ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Hertha Ayrton Britain (born in Portsea, England), 1854–1923

Hertha Ayrton (England, 1854 to 1923) was a scientist and an inventor. She studied the ripple patterns that water makes in sand, using a model sea in a zinc bath. She saw that the ripples came in matching pairs.

Try it (if you like): Ripples on a beach repeat: ridge, dip, dip, dip, then ridge, dip, dip, dip again. Is ripple number 20 a ridge or a dip? What about ripple number 21?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Look at the shape pattern in the Start here page again (places 1 to 10). How many circles are there in the first 30 shapes? How do you know?



1 2 3 4 5 6 7 8 9 10

place

No.2

A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?

No.3

Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?

No.4

Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?

Go further

No.5

Ali counts in 3s from 3. Mei counts in 4s from 4. They both say 12. Predict the next number they will BOTH say. Then list every number up to 100 that they both say. What do you notice? Why?

No.6

Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes? Is there a tower with exactly 50 cubes? With exactly 100 cubes? Explain how you know.

No.7

Frog A starts at 0 and jumps 4 each time: 4, 8, 12 ... Frog B starts at 2 and jumps 3 each time: 5, 8, 11 ... Which numbers up to 30 do both frogs land on?

START HERE. After question 3 you may choose: carry on, or try Think harder.

MYSTERY NUMBER MACHINES

Name _____ Group _____ Date _____

IN = what goes in. OUT = what comes out. Undo = do the opposite.

No.1

Mystery machines. Find the number that went IN and write it in the yellow box. Undo the LAST step first, then check forwards.

a I multiply by 2, then add 7. I get 27.

?	→	× 2	→	+ 7	→	**27**

b I take away 8, then multiply by 5. I get 40.

?	→	− 8	→	× 5	→	**40**

c I divide by 3, then add 12. I get 20.

?	→	÷ 3	→	+ 12	→	**20**

d I multiply by 4, then take away 15. I get 21.

?	→	× 4	→	− 15	→	**21**

No.2

Two-step number machines.

a This machine does × 3, then + 2. Fill in the table.

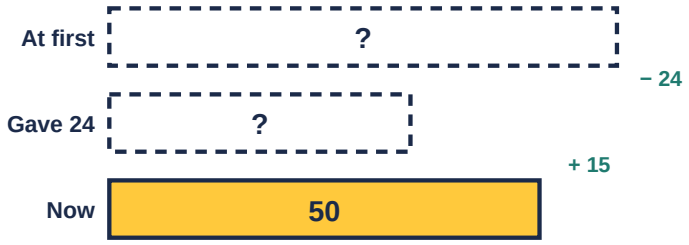
IN	4	10		
OUT			17	29

b This machine does × 2, then + 1. Fill in the table for IN 3, 5, 10 and 20.

IN	3	5	10	20
OUT				

No.3

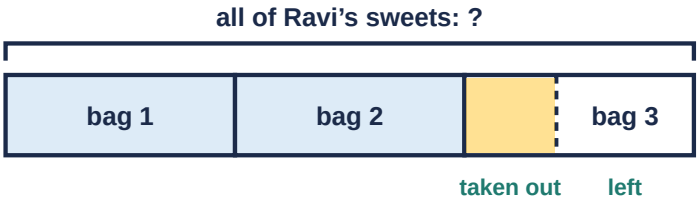
Siti had some stickers. She gave 24 stickers to her brother. Then her mother gave her 15 more. Now she has 50 stickers. How many stickers did Siti have at first? Fill in the two ? bars first (start from what Siti has now).



Answer: _____

No.4

Ravi put all his sweets into 3 bags, with the same number in each bag. He took 4 sweets out of bag 3. Now bag 3 has 6 sweets. How many sweets did Ravi have altogether?



Answer: _____

No.5

Mei and Ali both put 5 into a machine.

a Mei's machine does + 4, then × 2. What comes out? _____

b Ali's machine does × 2, then + 4. What comes out? _____

c Do the two machines always give the same answer? _____

d What must Ali put IN to get the same OUT as Mei? _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

A magic trick. Think of a number. Add 5. Double it. Take away 10. Halve it. (a) Try 5 different starting numbers. What do you notice about your final answer? (b) Is it ALWAYS true? Try a big number and a small number, then say why. (c) Make your own trick that always gives you back the number you started with. Test it on 3 numbers.

Easy first step: do the trick with 3, then with 10, and write down each step.

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Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ drew the machine ☐ worked backwards ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Mary Somerville Scotland (and later Italy), 1780–1872

Mary Somerville grew up in Scotland about 240 years ago. She told how a magazine puzzle full of X and Y made her want to learn algebra, so she taught herself from books (this comes from her own memories, so the details may not be exact).

Try it (if you like): Mary's X was a mystery number. I double X, then take away 3, and the answer is 9. Work backwards to find X.

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

A machine multiplies by a number, then adds a number. IN 2 gives OUT 11 and IN 5 gives OUT 20. Find what the machine does (try × 2, × 3, × 4 ... with IN 2, then test IN 5). Then: what OUT does IN 10 give? Which IN gives OUT 26?

No.2

Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?

No.3

Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

No.4

Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.

Go further

No.5

Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?

No.6

A machine does × 2, then – 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?

No.7

Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?

START HERE. After question 3 you may choose: carry on, or try Think harder.

MARKET DAY

Name _____ Group _____ Date _____

Coins: 5¢, 10¢, 20¢, 50¢, $1. Notes: $2, $5, $10, $50, $100. (There is no $20 note.) 100¢ = $1.
Careful: 50¢ is a coin, $50 is a note.

No.1

Pay exactly. Use the FEWEST notes and coins you can.

Item	Price	Notes and coins I use	Pieces
Curry puff	$1.65		
Notebook	$4.30		
Umbrella	$18.75		
Bicycle	$76		

No.2

Find EVERY way to pay exactly $8 using $5 notes, $2 notes and $1 coins. Make a list in order: start with the most $5 notes. You may not need every row.

Way	Number of $5	Number of $2	Number of $1	Pieces altogether
A				
B				
C				
D				
E				
F				
G				
H				

a How many ways are there? _____

b Two ways use exactly 4 pieces of money. What are they?

Answer: _____

No.3

Change.

a Mei buys an eraser for 65¢. She pays with a $1 coin. How much change does she get?

_____

b Ali buys a game for $26. He pays with a $50 note. How much change does he get? Which notes give this change with the fewest notes?

Answer: _____

No.4

Siti has $20 to spend.



a Can Siti buy the bag and the cap? Explain.

b Siti buys two different things and gets exactly $3 change. Find ALL the pairs she could buy.

No.5

Work backwards.

a Mei spent 45¢ on a sweet and 30¢ on a pencil. She has 25¢ left. How much money did she have at first? Write it in dollars and cents. _____

No.6

Stickers cost 20¢ each. Erasers cost 50¢ each. Siti spends exactly $2. How many stickers and erasers could she buy? Find ALL the ways. (She might buy only one kind.)

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Most coins. Use only 5¢, 10¢, 20¢ and 50¢ coins. (a) Pay 35¢ using the fewest coins. How many coins is that? (b) Try the other amounts from 5¢ to 95¢ (in steps of 5¢) and make a table of how many coins each needs. (c) Which amount needs the MOST coins, even when you use the fewest possible? *Needs: play coins (5¢, 10¢, 20¢, 50¢), or paper and a pencil.*

Easy first step: pay 35¢ with the fewest coins from the 5¢, 10¢, 20¢ and 50¢ coins.

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ used play coins ☐ made a list in order ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Luca Pacioli Italy, about 1447 – 1517 (sources differ)

Luca Pacioli was a maths teacher in Italy about 500 years ago. In 1494 he printed a big maths book that also showed how a shop keeps its money records. He did not invent the method: he wrote down what Venice merchants already did.

Try it (if you like): Keep a record like a merchant. You have $20. You spend $6 on a book and $5 on a cap. How much did you spend altogether, and how much is left?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.

No.2

How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

No.3

Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

No.4

How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.

Go further

No.5

Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?

No.6

Sweets cost 5¢ each and lollipops cost 20¢ each. Ben buys 8 things altogether and spends exactly $1. How many sweets and how many lollipops does he buy?

No.7

Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?

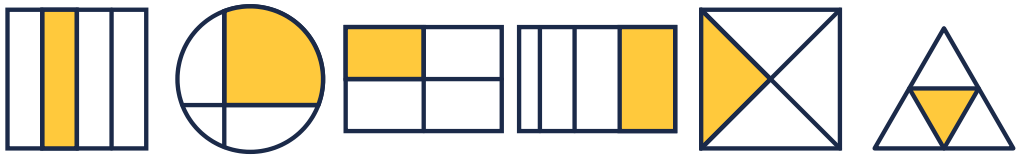
START HERE. After question 3 you may choose: carry on, or try Think harder.

FAIR SHARES

Name _____ Group _____ Date _____

No.1

Fair shares? Each shape has 4 parts and 1 part is shaded. Circle the shapes that are cut into quarters (4 EQUAL parts).



A B C D E F

a Choose one shape that is NOT cut into quarters. Explain how you know.

.....

No.2

What fraction is shaded? What fraction is NOT shaded? Write them in the boxes.

a



shaded ☐ not shaded ☐

b



shaded ☐ not shaded ☐

c



shaded ☐ not shaded ☐

No.3

Compare the fractions. Draw bars the same length if it helps.

a Circle the bigger fraction in each pair: $\frac{1}{3}$ or $\frac{1}{8}$ $\frac{5}{9}$ or $\frac{7}{9}$

b Smallest first: $\frac{1}{5}$ $\frac{1}{2}$ $\frac{1}{10}$ $\frac{1}{4}$ _____ _____ _____ _____

c Mei says: "$\frac{1}{6}$ is bigger than $\frac{1}{4}$, because 6 is bigger than 4." Is she right? Explain.

.....

No.4

Add or take away. Colour the bars to help you. Write each answer in the box.

a 

$\frac{2}{5} + \frac{1}{5} = \square$

b 

$\frac{3}{8} + \frac{4}{8} = \square$

c 

$\frac{4}{12} + \frac{5}{12} = \square$

d 

$\frac{1}{6} + \square = 1 \text{ whole}$

e 

$\frac{7}{10} - \frac{3}{10} = \square$

No.5

Mum cuts a cake into 8 equal slices. Ali eats 2 slices. Mei eats 3 slices.

Cake 

a What fraction of the cake do they eat altogether? _____

b What fraction of the cake is left? _____

c Who eats more? How much more? _____

No.6

Ravi and Siti each have a pizza. The pizzas are the same size. Ravi cuts his into 4 equal slices and eats 1 slice. Siti cuts hers into 8 equal slices and eats 3 slices. Who eats more pizza? Hint: fold or line up the bars; one quarter covers two eighths. Colour the bars to show how you know.

Ravi 

Siti 

Who eats more? _____

.....

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Fraction strips. (a) Cover a HALF strip exactly with pieces of ONE kind only. Which kinds work? How many pieces of each? (b) Do the same for a THIRD strip and a QUARTER strip. (c) Can thirds ever cover a half exactly? Why not? *Needs: the fraction strips from the cut-out page (halves, thirds, quarters, sixths, eighths).*

Easy first step: lay the two quarter strips on top of the half strip. Do they fit exactly?

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ folded or drew strips ☐ drew bars ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Ahmes the scribe Egypt, about 1550 BC (date on the papyrus copy; sources differ)

Ahmes was a scribe in ancient Egypt more than 3,500 years ago. He wrote out a long maths roll on papyrus with 84 problems, including fractions like 1/3 and 1/4. It is now in the British Museum in London.

Try it (if you like): Share 1 loaf of bread equally among 4 children. What fraction of the loaf does each child get? What fraction do 3 of the children get together?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

No.2

A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?

No.3

2/7 + ? = 1 whole

No.4

Ali says: "Half of my pizza is bigger than half of your pizza." Can he be right? Explain.

Go further

No.5

A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?

No.6

A paper strip is folded in half, then in half again, then in half again. When you open it, how many equal parts are there? What fraction is each part?

No.7

Siti eats 1/2 of a pizza. Then Ravi eats 1/2 of what is left. What fraction of the whole pizza is left now?

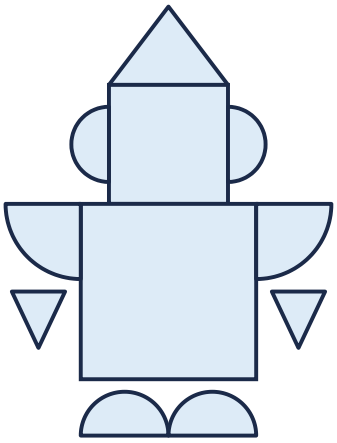
START HERE. After question 3 you may choose: carry on, or try Think harder.

SHAPE BUILDERS

Name _____ Group _____ Date _____

No.1

The robot is made of pieces. Count each kind of piece and write how many in the boxes. Tick each piece as you count it.



Square	
Triangle	
Semicircle	
Quarter circle	

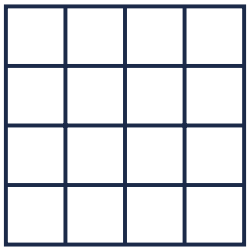
a Which kind is there the most of? _____ b Pieces altogether? _____

c Mei has 3 semicircles and 6 quarter circles, all cut from same-size circles. How many whole circles can she make? Show how.

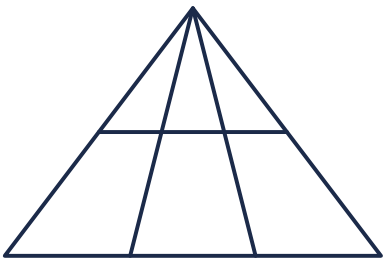
No.2

Count every shape. Work in order. In (a), colour each 2 by 2 square you find a different colour.

a



b

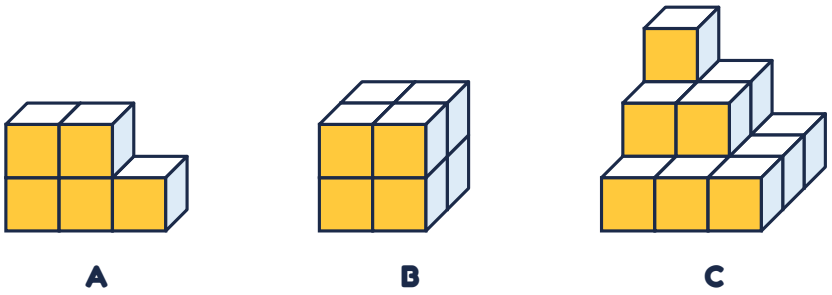


a How many squares?	Number
1 by 1 squares	
2 by 2 squares	
3 by 3 squares	
4 by 4 squares	
Squares altogether	

b Triangles on the bottom line: _____ on the line across: _____ altogether: _____

No.3

Count the cubes. Build stack A with your cubes first, then count. Every cube is on the table or sits on top of another cube. Stack A is only 1 cube deep.



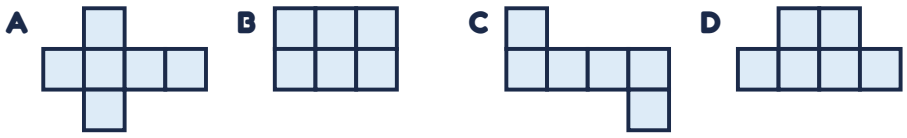
A B C

a A: _____ cubes B: _____ cubes C: _____ cubes

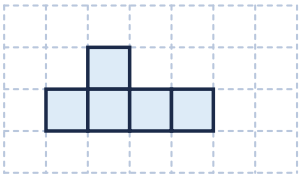
b Ravi wants to turn stack C into a big cube 3 cubes long, 3 cubes wide and 3 cubes high. How many more cubes does he need? _____

No.4

Cut out nets A to D from the cut-out page. Fold net A with your tutor on camera first, then fold B, C and D. Circle the letters of the nets that fold up into a cube.



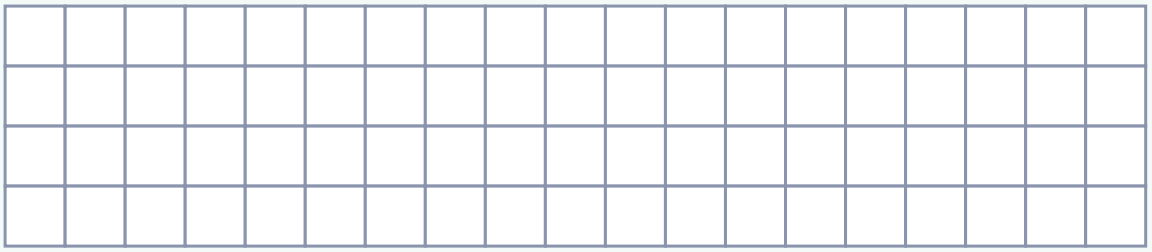
b Add ONE more square to this shape so that it folds into a cube. Draw it on the grid.



Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Squares joined edge to edge. Squares must touch along a whole side. A shape that is turned round or flipped over counts as the SAME shape. (a) Find ALL the different shapes you can make with 3 squares. (b) Now find ALL the different shapes you can make with 4 squares. Draw each one on the grid. (c) How do you know you have found them all? *Needs: square tiles or small squares of paper, and the grid on this page.*

Easy first step: join 2 squares edge to edge. How many different shapes can you make? Then add a third square.



.....

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ used cubes or tiles ☐ drew it ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Alicia Boole Stott Ireland (born in Cork; later lived in England), 1860–1940

Alicia Boole Stott was born in Cork, Ireland, in 1860. She never went to university, but she learned from her mother and from geometry books. Later she cut and joined cardboard to build models of very unusual shapes.

Try it (if you like): A cube is built from flat square pieces. How many square faces does a cube have? How many edges does it have?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.

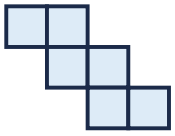
No.2

How many squares are in a grid 3 squares long and 2 squares high?

No.3

Cut out nets E and F from the cut-out page and try folding them. Which one folds up into a cube? How do you know?

E



F



No.4

Six squares are joined in a straight line. Can they fold into a cube? Explain.

Go further

No.5

A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?

No.6

How many squares can you find in a grid 5 squares long and 5 squares high?

No.7

On a dice, opposite faces add up to 7. You can see 3 faces showing 1, 2 and 3. What numbers are on the 3 faces you cannot see? What do they add up to?

START HERE. After question 3 you may choose: carry on, or try Think harder.

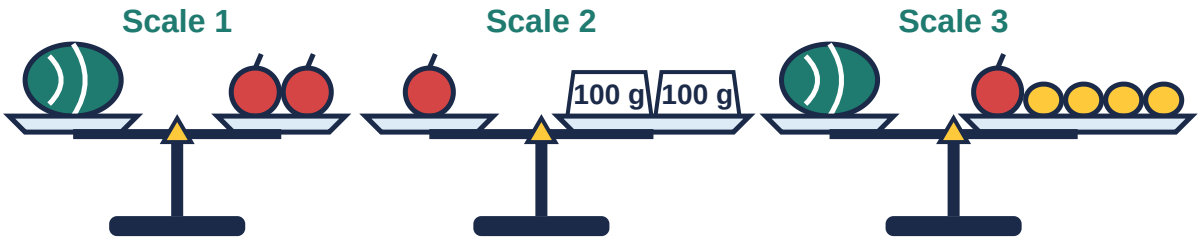
MEASURE MASTERS

Name _____ Group _____ Date _____

A level balance means both sides are equally heavy. Show how you know for every answer.

No.1

Each scale is level.



a How heavy is 1 apple? _____ g

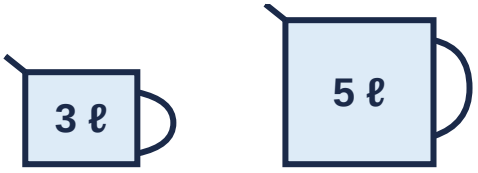
b How heavy is the melon? _____ g

c How heavy is 1 lemon? Show how you know. Tip: on Scale 3, melon = apple + 4 lemons.

d How many lemons balance the melon? _____ lemons

No.2

You have a 3 ℓ jug and a 5 ℓ jug. You can fill a jug from the tap, empty a jug into the sink, or pour from one jug into the other until one jug is full or the other is empty.



No marks on the jugs!

a How can you get exactly 2 ℓ? Write the steps.

b How can you get exactly 1 ℓ? Write the steps.

No.3

Next puzzle. Start with both jugs empty. The first two rows are filled in for you. Now get exactly 4 ℓ in the 5 ℓ jug. Write each step in the table. You may not need every row.

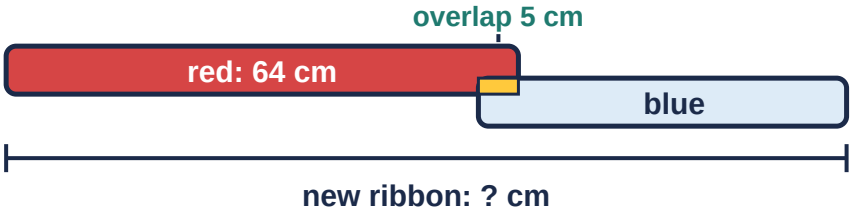
Step	What I do	3 ℓ jug	5 ℓ jug
Start	both jugs empty	0 ℓ	0 ℓ
1	Fill the 5 ℓ jug	0 ℓ	5 ℓ
2			
3			
4			
5			
6			

No.4

Siti has a red ribbon 64 cm long. Her blue ribbon is 18 cm shorter than the red ribbon.

a How long is the blue ribbon? _____ cm

b She glues them end to end with a 5 cm overlap. The new ribbon is _____ cm long.



No.5

Read the clocks. Hint: on clock B the hour hand is almost on the 5. Is it 4 o'clock or 5 o'clock?

A



Ali is eating breakfast.

B



Ali is at swimming after school.

short thick hand = hour long red hand = minute

a Write each time with a.m. or p.m. A: _____ B: _____

b Swimming starts at the time on clock B and lasts 35 minutes. It ends at _____

c Mei's family leaves home at 10:45 a.m. They reach Grandma's flat at 11:30 a.m. How long is the trip? _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Weights puzzle. You have three different weights. Each weight is 1 g, 2 g, 3 g, 4 g or 5 g. You put some of them on one pan to make a mass. (a) With weights of 1 g, 2 g and 4 g, which masses from 1 g to 7 g can you make? (b) Find ALL the sets of three different weights from 1 g to 5 g that can make every mass from 1 g to 6 g. (c) What do all the good sets have in common? *Needs: paper and a pencil (cubes or coins can stand for the weights).*

Easy first step: with weights of 1 g, 2 g and 4 g, can you make 3 g? Can you make 5 g?

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ made a table ☐ used cups or objects ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Thales of Miletus Miletus, a Greek city (now in Turkey), about 624 – about 547 BC

Thales lived in the Greek city of Miletus, now in Turkey, about 2,600 years ago. A story told by writers who lived long after him says he measured a pyramid using its shadow (historians cannot be sure it happened like this).

Try it (if you like): At the moment your shadow is as long as you are tall, a flagpole's shadow is 9 m long. How tall is the flagpole? If your shadow is 120 cm long at that moment, how tall are you?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

Ben cuts a 20 m rope into 4 equal pieces. How long is each piece? How many cuts does he make?

No.2

Mei's family leaves home at 10:45 a.m. They reach Grandma's flat at 12:10 p.m. How long is the trip?

No.3

3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

No.4

You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.

Go further

No.5

You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.

No.6

Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?

No.7

Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 a.m. What time does the last lesson end?

START HERE. After question 3 you may choose: carry on, or try Think harder.

CHALLENGE DAY

Name _____ Group _____ Date _____

No.1

The picture graph shows the ice-creams a stall sold.

Monday	
Tuesday	
Wednesday	
Thursday	Draw the pictures for Thursday here.

 = 5 ice-creams

a How many ice-creams were sold on Tuesday? _____

b 90 ice-creams were sold in the 4 days. Draw Thursday's pictures on the graph.

c Each ice-cream costs $2. How much did the stall get on Wednesday? _____

No.2

Use the digit cards 3, 0 and 8 once each to make a 3-digit number.

a List ALL the numbers in order. _____

b Greatest: _____ Smallest: _____

c Which of them are between 300 and 810? _____

No.3

Ben has 6 coins. They are all 50¢ and 20¢ coins. Together they are worth $1.80. How many of each coin does he have? Make a table.

Number of 50¢	Number of 20¢	Total
0	6	$1.20

Ben has _____ 50¢ coins and _____ 20¢ coins.

No.4

Ali, Mei, Siti and Ben each go to school a different way: bus, MRT, car or walking. Put a tick where it fits and a cross where it cannot.

Clue 1: Siti lives next to the school, so she walks.

Clue 2: Ben does not go by bus or by car.

Clue 3: Mei does not go by car.

	Bus	MRT	Car	Walk
Ali				
Mei				
Siti				
Ben				

Ali: _____ Mei: _____ Siti: _____ Ben: _____

No.5

A cake is cut into 10 equal pieces. Ali eats 3/10 of the cake. Mei eats 2 more pieces than Ali. Ben eats the rest.



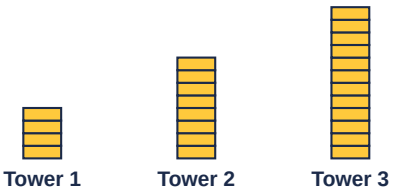
a What fraction of the cake does Mei eat? _____

b What fraction of the cake does Ben eat? _____

c Order the children, the one who eats the most first. _____

No.6

Ben builds towers with cubes. Tower 1 has 4 cubes, Tower 2 has 8 cubes and Tower 3 has 12 cubes. Keep the pattern going.



Tower	1	2	3	4	5	6
Cubes	4	8	12			

a How many cubes are in Tower 8? _____

b Which tower has 36 cubes? Tower _____

Investigation (a puzzle with more than one good answer. Try it any time you like, or at home.)

Staircase. Mei climbs a staircase. She can take 1 step or 2 steps at a time. For 3 steps there are 3 ways: 1+1+1, 1+2 and 2+1. (a) Find ALL the ways for 4 steps. (b) Find ALL the ways for 5 steps. (c) Fill in a table for 1, 2, 3, 4 and 5 steps. What pattern do you see? Can you guess the number of ways for 6 steps?
Needs: paper and a pencil.

Easy first step: draw 3 steps and show the 3 ways: 1+1+1, 1+2 and 2+1.

Looking back (2 minutes. You may pass on any line.)

Something I did well: .....

The hardest part was ... .

What helped me: ☐ listed in order ☐ drew it ☐ I asked for help ☐ I am still stuck (my tutor will read this at the next lesson)

Story corner: Paul Erdos Hungary, 1913–1996

Paul Erdos was born in Budapest, Hungary, in 1913. He loved maths problems and loved working on them with other people: he worked with more than 500 different mathematicians. For him, a good problem was something to share.

Try it (if you like): Solve this with a partner. Two friends have 12 sweets between them. One has 4 more than the other. How many sweets does each friend have?

My answer: .....

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder

No.1

30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?

No.2

How many 3-digit numbers have digits that add up to 3? List them all.

No.3

A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.

No.4

5, 9, 13, 17, ... What is the 10th number in this pattern?

Go further

No.5

A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?

No.6

Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?

No.7

At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy? What is the smallest total 7 dishes can cost? The biggest? Can 7 dishes cost $30? Can they cost EVERY whole-dollar amount in between?

Cut-outs

For the Level 2 program. Cut along the dashed lines and keep the pieces in one envelope.

Place-value cards (Lessons 1, 2 and 10)

100	10	1
200	20	2
300	30	3
400	40	4
500	50	5
600	60	6
700	70	7
800	80	8
900	90	9

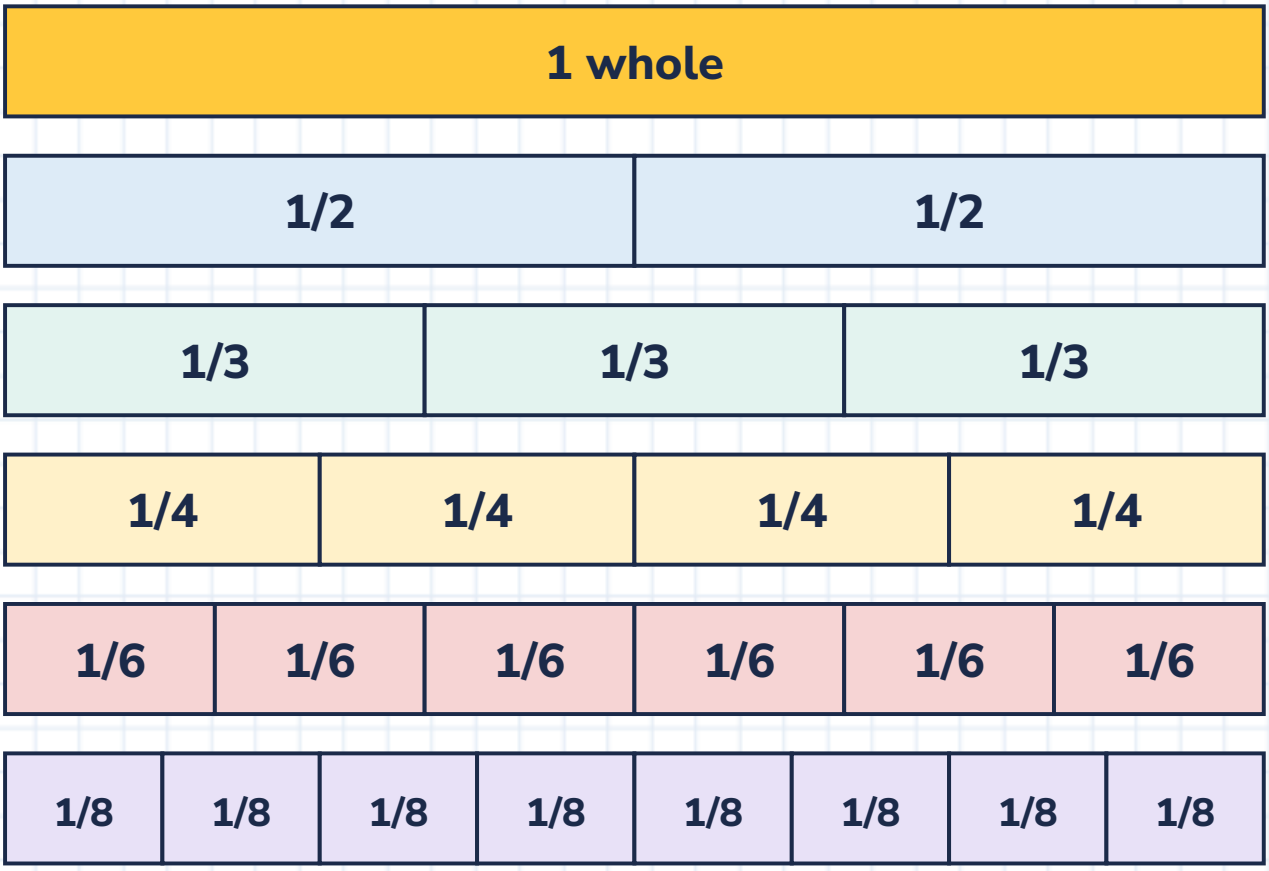
Lay the cards on top of each other, right edges together: 300 + 40 + 7 shows 347.

Digit cards

Two sets of 0 to 9 (Lessons 1 and 10).

0	1	2	3	4
5	6	7	8	9
0	1	2	3	4
5	6	7	8	9

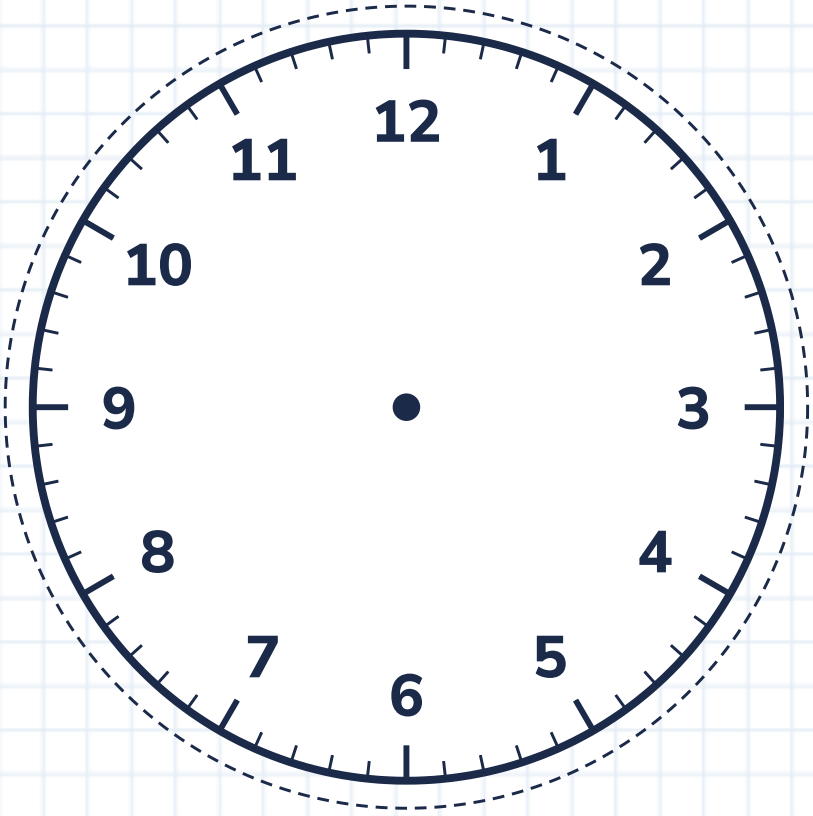
Fraction strips (Lesson 7)

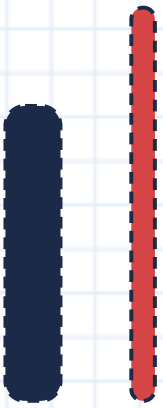


Cut along each strip, then cut the parts apart. Compare: which is bigger, 1/3 or 1/4?

Paper clock

For Lesson 9 (time).





hour hand (short, thick)

minute hand (long, thin)

Cut out the clock and the hands. Pin them at the centre with a split pin or a short pencil.

Paper strips of numbered squares (Lesson 9: ribbons and lengths)

Strip A

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

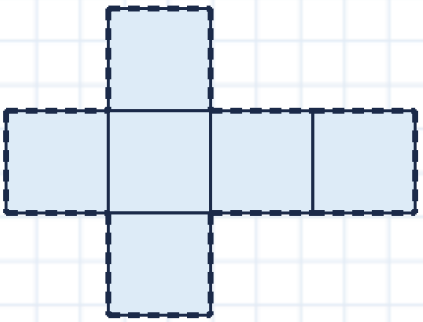
Strip B

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

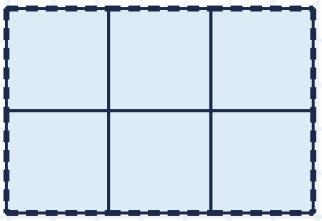
Cube nets

For Lesson 8 (Shape Builders). Cut out each net and try folding it. Which ones make a cube?

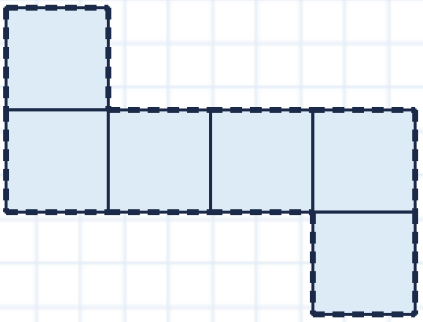
Net A



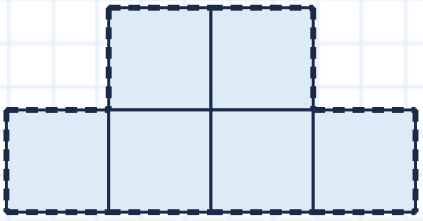
Net B



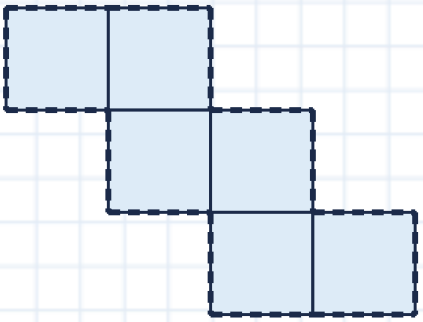
Net C



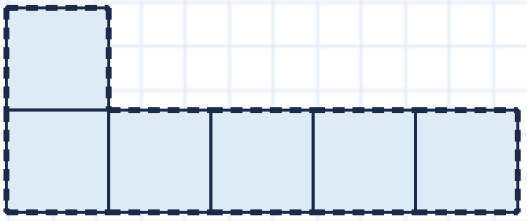
Net D



Net E



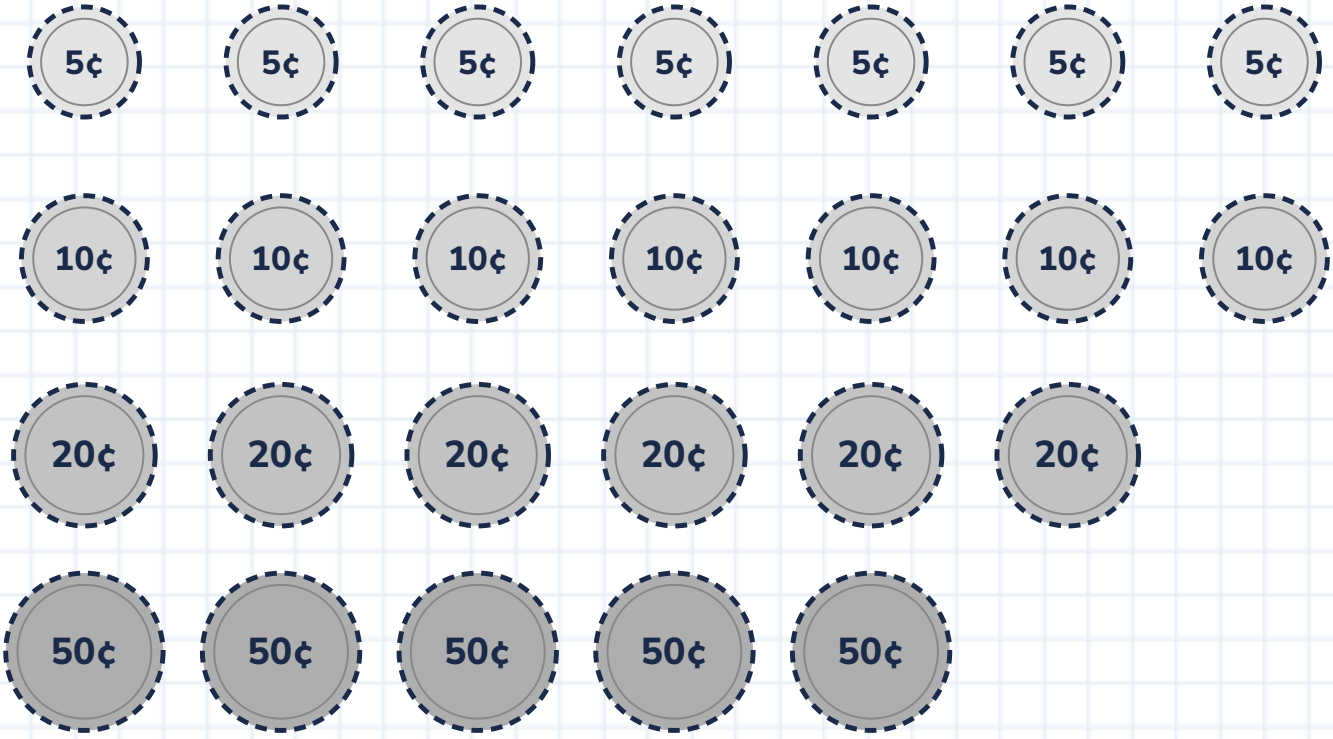
Net F



Cut along the dashed lines. Fold along the solid lines. Which nets fold up into a cube?

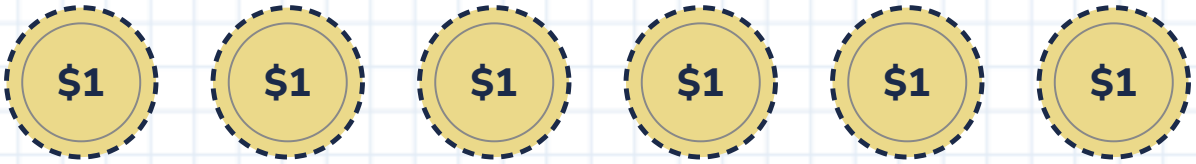
Play coins

For Lesson 6 (Market Day). Seven 5¢, seven 10¢, six 20¢ and five 50¢. Keep spare coins in the envelope.



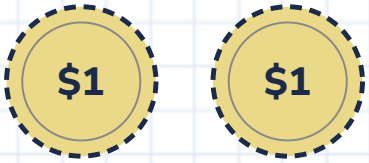
Play notes and $1 coins

For Lesson 6 (Market Day). Eight $1 coins (six here and two spare below), four $2, four $5, four $10 and two $50 notes. Keep spare coins in the envelope.





Two spare $1 coins



CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Level 2 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____